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Question **1**

Complete

Not graded

Evaluate the following as true or false.

The infinite series $\sum_{n=1}^{\infty} \frac{1}{2n+1}$ converges.

Select one:

- True
- False

$$\int_1^{\infty} \frac{1}{2x+1}dx = \lim_{a\to\infty} \ln|2x+1|\Big|_1^a$$
$$= \lim_{a\to\infty} \left(\ln|2a+1| - \ln 3\right)$$
$$= \infty$$

Therefore, the series

$\sum_{n=1}^{\infty}\left[1/(2n+1)\right]$ diverges.

The correct answer is 'False'.

Question **2**

Complete

Not graded

Consider the series $\sum_{n=1}^{\infty}\left(-\frac{1}{n^2}\right)$.

Can you use the integral test to judge its convergence?

Select one:

- a. yes
- b. no

Because $\int_1^{\infty}\frac{1}{x^2}dx$ converges,

$\sum_{n=1}^{\infty}\frac{1}{n^2}$ converges. Subsequently,

the series $\sum_{n=1}^{\infty}\left(-\frac{1}{n^2}\right)=-1\sum_{n=1}^{\infty}\frac{1}{n^2}$

converges.

The correct answer is: yes

Question **3**

Complete

Not graded

Suppose that $f(x)$ is a continuous, positive and decreasing function on $[0, \infty)$ with $a_n = f(n)$.
Evaluate the following as true or false.

If $\sum_{n=1}^{\infty} a_n$ does not converge, then the integral $\int_1^{\infty} f(x)\,dx$ does not converge.

Select one:

- True
- False

If the integral $\int_1^{\infty} f(x)$ converges, then the series $\sum_{n=1}^{\infty} a_n$ must converge, which is a contradiction to the assumption.

The correct answer is 'True'.

Question **4**

Complete

Not graded

Evaluate the following as true or false.

The series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}+1}$ converges.

Select one:

- True
- False

Let $u = \sqrt{x}$, then $du = \frac{1}{2}x^{-1/2}dx$,

or $dx = 2u\,du$.

$$\int_1^{\infty} \frac{1}{\sqrt{x}+1}dx = \int_1^{\infty} \frac{2u}{u+1}du$$

$$= \int_1^{\infty} \left(2 - \frac{2}{u+1}\right)du$$

$$= \lim_{a\to\infty} 2u - 2\ln|u+1| - [2 - 2\ln(2)]$$

$$= \lim_{a\to\infty} \ln\left[\frac{e^{2u}}{(u+1)}\right] - 2 + 2\ln(2)$$

$$= \infty$$

Therefore the series diverges.

The correct answer is 'False'.

Question **5**

Complete

Not graded

Evaluate the following as true or false.

The series $\sum_{n=1}^{\infty} \frac{1}{n^{1.01}}$ converges.

Select one:

- True
- False

This is a *p*-series with $p = 1.01$. Because $1.01 > 1$, the series converges.

The correct answer is 'True'.

Question **6**

Complete

Not graded

Evaluate the following as true or false.

The series $1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} + \ldots$ is convergent.

Select one:

- True
- False

The series $1+\frac{1}{8}+\frac{1}{27}+\frac{1}{64}+\cdots$

can be rewritten as $\sum_{n=1}^{\infty}\frac{1}{n^3}$.

This is a p-series, with $p > 1$,

so the series converges.

The correct answer is 'True'.

Question **7**

Complete

Not graded

The series $\sum_{n=1}^{\infty} \frac{(\ln\ n)^2}{n^2}$ converges.

Select one:

- True
- False

$$\int_1^{\infty} \frac{(\ln x)^2}{x^2} dx = -\frac{1}{x}(\ln x)^2\Big|_1^{\infty} + \int_1^{\infty} \frac{1}{x} 2(\ln x)\frac{1}{x} dx$$

$$= 2\int_1^{\infty} \frac{\ln x}{x^2} dx$$

$$= -2\frac{1}{x}(\ln x)\Big|_1^{\infty} + 2\int_1^{\infty} \frac{1}{x^2} dx$$

$$= -2\frac{1}{x}\Big|_1^{\infty} = 2$$

Therefore, the series $\sum_{n=1}^{\infty} \frac{(\ln n)^2}{n^2}$ converges.

The correct answer is 'True'.

Question **8**

Complete

Not graded

Suppose that $f(x)$ is a continuous, positive and decreasing function on $[\,0,\,\infty)$ with $a_n = f(n)$.

Evaluate the following as true or false.

Assuming that $\int_1^{\infty} f(x)\,dx$ converges, $\sum_{n=500}^{\infty} a_n$ converges.

Select one:

- True
- False

The series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=500}^{\infty} a_n$ only differ by a constant, namely $C = \sum_{n=1}^{499} a_n$. The integral test states that $\sum_{n=1}^{\infty} a_n$ converges, subsequently the series $\sum_{n=500}^{\infty} a_n$ converges. In fact, if $\sum_{n=1}^{\infty} a_n$ converges to L then $\sum_{n=500}^{\infty} a_n$ converges to $L - C$.

The correct answer is 'True'.

Question **9**

Complete

Not graded

Suppose that $f(x)$ is a continuous, positive and decreasing function on $[\,0, \infty)$ with $a_n = f(n)$.

The series $\sum_{n=1}^{\infty} a_n$ converges provided that

Select one:

- a. $\int_1^{\infty} f(x)\,dx$ diverges.
- b. $\int_1^{2000} f(x)\,dx = 0.$
- c. $\int_{2000}^{\infty} f(x)\,dx$ converges.
- d. $\lim_{n\to\infty} f(x) = 0.$

Because $\int_{2000}^{\infty} f(x)dx$ converges, $\int_1^{\infty} f(x)dx$ converges (because they only differ by a constant), and then the integral test implies that the series is convergant.

The correct answer is: $\int_{2000}^{\infty} f(x)\,dx$ converges.

Question **10**

Complete

Not graded

Evaluate the following as true or false.

The series $\sum_{n=1}^{\infty} \frac{1}{n^{0.99}}$ converges.

Select one:

- True
- False

This is a *p*-series with $p = 0.99$. Because $0.99 < 1$, the series is divergent.

The correct answer is 'False'.

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